MILITARY SPECIFICATION
REMOVER, AIRCRAFT CANOPY, M8A2
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, M8A2, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-601	-	Aluminum Alloy Sand Castings

Military

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-C-6021	-	Castings, Classification and Inspection
MIL-A-21180	-	Aluminum Alloy Casting, High Strength
MIL-C-46227	-	Components for Propellant Actuated Devices, General Specification for

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Inspection Terms and Definitions
MIL-STD-413	-	Visual Inspection Guide for Rubber O-Rings

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:02Z
Check the source to verify that this is the current version before use.

DRAWINGS

Munitions Command

C8837890	-	Carton, Packing, Ammunition for Remover, Aircraft Canopy M8A1 or M8A2 with Cartridge M150
C9240994	-	Carton, Alternative Packing, Ammunition for (1) Remover, Aircraft Canopy, M8A1 or M8A2 with Cartridge M150
C9230161	-	Box, Packing, Ammunition for (1) Remover, Aircraft Canopy, M8A1 or M8A2, with Cartridge M150
LI-10522143	-	Inspection Equipment List for Remover, Aircraft Canopy M8A2 Metal Parts Assembly
F-10522143	-	Remover, Aircraft Canopy, M8A2 Metal Parts Assembly

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018.

3. REQUIREMENTS

3.1 General.-Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, reference specifications, and the following:

3.2 Material.-The contractor shall be responsible for selecting from the classification list contained in the raw material specifications, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Physical Properties.-Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

Source: https://assist.dla.mil -- Downloaded: 20260-23T01:02Z
Check the source to verify that this is the current version before use.

3.4 Hydrostatic Pressure.-The outside tube, D10522060, and inside tube, D10522036, shall withstand a hydrostatic pressure of 2500 pounds per square inch (psi) minimum, for 15 seconds, minimum, without deformation or mechanical failure (see 6.2).

3.5 Manual Function.-Without electrical power, and with the clutch disengaged, the pull force required to extend the inside tube assembly shall not exceed 40 pounds.

3.6 Reversibility Load.-The remover shall be capable of withstanding a load of 2500 pounds minimum in tension and 1740 pounds minimum in compression at a mid-stroke of approximately 5 inches. The remover shall not travel or unlock due to the applied load.

3.7 Operating Performance.-The remover shall be cycled and shall be capable of operating to the loads described by the load curves A, B, and C of Figure 1 (see 6.3) and shall comply with the following:

3.7.1 The motor running current drain under load curve B of Figure 1 shall not exceed 13 amperes at 26 to 30 volts D.C. at ambient conditions.

3.7.2 The motor running current drain under load curve C of Figure 1 shall not exceed 20 amperes at 26 to 30 volts D.C. at ambient conditions.

3.7.3 Time required for linear extension of the remover for a normal stroke of 9.38 inches and under load curve B of Figure 1 shall not exceed 12 seconds using the power requirements specified in 3.5.1.

3.7.4 Time required for linear extension of the remover for a normal stroke of 9.38 inches and under curve C of Figure 1 shall not exceed 15 seconds using the power requirement specified in 3.7.2.

3.7.5 Time requirement for linear retraction of the remover for a normal stroke of 9.38 inches under curve A of Figure 1 shall not exceed 15 seconds at 26 to 30 volts D.C. with a running current drain not to exceed 15 amperes at ambient conditions.

3.7.6 When the load aids the operation of the remover for the full stroke of 9.38 inches, the operating time shall be 5 seconds minimum to 10 seconds maximum and the running current shall not exceed 10 amperes at 26 to 30 volts D.C.

3.7.7 There shall be no structural failure of any component under the inertia loads imposed by the sudden interruption of electrical power.

3.7.8 At the end of the duty cycle for normal load curve B, Figure 1, the remover shall pull against the stop with a force, known as cinch-down load, of 1000 to 1450 pounds. Under this cinch-down load requirements, the load limit switch shall de-energize the remover electic motor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

3.7.9 Time required to engage clutch shall not exceed 500 milliseconds (see 6.4).

3.7.10 After the remover has cinched down, the clutch shall be disengaged. The time required to disengage the clutch shall not exceed 500 milliseconds (see 6.5).

3.7.11 The remover shall not fail to operate under normal load curve B of Figure 1 at a voltage input of 18 volts D.C.

3.8 Metal Parts Assembly.-Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing.

3.9 Protective Finish.-Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.10 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, reference specifications and the following:

3.10.1 Metal Defects.-Components shall be free from cracks, splits, or other metal defects.

3.10.2 Burr.-No part shall have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.10.3 Foreign Matter.-Chips, dirt, grease (except where required), rust, corrosion, or other foreign matter shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 5 assemblies, 10 sets of unassembled parts and 50 shear pins from the same lot of shear pins tested by the contractor. The assemblies and shear pins shall be subjected to the tests in 4.4. The sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Component and Subassembly Inspection shall be in accordance with MIL-C-46227, Category A, except as specified herein.

4.3.1.3 Size of lot.-For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 removers, maximum, exclusive of test samples, provided the homogeneity requirements of manufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein using the sampling plan and associated criteria in Table I and II. For an individual defect (single characteristic), the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The roughness comparison specimens prescribed

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:02Z
Check the source to verify that this is the currentversion before use.

4.3.2 (Cont'd)

in ASA B46.1, shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE II

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

Source: https://assist.dla.mil -- Downloaded: 202610-23T01:02Z
Check the source to verify that this is the current version before use.

4.3.2.1 Metal Parts Assembly, Drawing F10522143.

Defects	Method of Inspection
CRITICAL	
1. Length of firing pin striker	Gage
2. Improper or insecure assembly	Manual & Visual
3. Loose or improper wiring	Manual & Visual
4. Improper soldering	Visual
5. Gas port hole obstructed	Visual
6. Any part missing	Visual
7. Marking missing, incorrect or illegible	Visual
8. Burr	Visual
9. Foreign matter	Visual
MAJOR	
None defined	
MINOR	
None defined	

4.3.2.2 Packing in Inner Box, Drawing C8837890.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Top of box loose	Manual
203. Strapping, missing, loose or improperly assembled	Manual & Visual
204. Supports or fillers missing from box	Visual
205. Box damaged	Visual
206. Seal missing or improperly positioned	Visual
207. Motor wires loose	Visual
208. Marking missing, incorrect or illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.3.2.3 Carton, Packing, Alternative C9240994.

Defects	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Carton broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect or illegible	Visual
205. Tape incomplete	Visual

4.3.2.4 Packing in Outer Box, Drawing C9230161.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Liner, fillers, cushings or spacers missing	Visual
203. Strapping missing, loose or improperly assembled	Visual
204. Hardware missing, loose or improperly engaged	Visual
205. Box damaged	Visual
206. Contents of box exposed	Visual
207. Car seal, missing or improperly sealed	Visual
208. Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with 4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

4.3.3.1 Hardness.-Fifty units from each heat treated lot of components shall be tested for compliance with drawing requirement. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Gear box tension test.-Sample size and acceptance criteria shall be in accordance with MIL-A-21180. Preparation of Gear box casting for tension test shall be as specified in 4.4.3.1 through 4.4.3.1.1.

4.3.3.3 Shear pin strength.-Fifty pins from each lot shall be tested for compliance with the values noted in the applicable drawing. Failure of any pin to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.4 Hydrostatic pressure.-Each inside and outside tube shall be subjected to the hydrostatic pressure test. Failure of any item to comply with the requirements of 3.4 shall be cause for rejection of the item.

4.3.3.5 Protective finish.-The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standards listed on the drawing.

4.3.3.6 Manual function.-Each remover shall be tested for the pull force required to extend the inside tube. Failure of any remover to comply with the requirements of 3.5 shall be cause for rejection of the remover.

4.3.3.7 Reversibility load.-Each remover shall be tested for reversibility load. Failure of any remover to comply with the requirement of 3.6 shall be cause for rejection of the remover.

4.3.3.8 Operating load (Curve B).-Each remover shall be cycled electrically 5 times to the load curve B of Figure I. Failure of any remover to comply with the requirements of 3.7 shall be cause for rejection of the remover.

4.3.3.8.1 Operating load (Curves A and C).-Each remover that has been tested and accepted for curve B shall be cycled electrically 5 times each to the load curves A and C of Figure I. Failure of any remover to comply with the requirements of 3.7 shall be cause for rejection of the remover.

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-10522143.

4.4 Test methods and procedures.

4.4.1 Shear pin strength.-After the application of protective finish, the pins shall be tested individually in a fixture in conjunction with a tensile testing machine. The specified load requirement shall be applied at a steady rate of 0.125 inches per minute.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:02Z
Check the source to verify that this is the current version before use.

4.4.2 Hardness test.-Prior to the application of protective finish, a hardness testing machine shall be used to determine compliance with drawing requirements.

4.4.3 Tension Testing of Gear Box Casting.-The test shall be performed in accordance with MIL-A-21180 and as specified herein.

4.4.3.1 Preparation of Casting-(Dimensions given are those shown on drawing F10522008).

4.4.3.1.1 Machine top face (face opposite lug that carries rod-end bearings) to 2.166 + .010 dimension (Section D-D).

4.4.3.1.2 Bore .6870 + .0005 diameter hole on same center as 2.7500 + .0005 diameter (shown in Section D-D) and 3/4" - 1/64" from edge of lug (Front View).

4.4.3.1.3 Machine faces of lug to .332 - .005 width to base of lug (Top view).

4.4.3.1.4 Machine face that receives 1/4 - UNF 28 thread tap to 1.620 - 010 dimension (Front View).

4.4.3.1.5 Drill .093 + .005 diameter hole and tap 1/4 - 28 UNF thread for grease fitting. (Front View).

4.4.3.2 Test Method.-Gear Box shall be tested without Bearing in place.

4.4.3.2.1 Utilizing a tensile machine, secure gear box mounting point to supporting rod by assembling a heat-treated drill rod through eye of clevis on testing machine and casting flange bore. The steel drill rod shall be heat-treated to a rockwell hardness ("C" scale) between the range of 40 to 45. The rod should be approximately 3 inches in length and .687 - .001 in diameter.

4.4.3.2.2 Gradually apply a tension load to the mounting flange through rod and flange base at the rate of 0.1 inch per minute until casting ruptures.

4.4.3.2.3 Load to cause rupture shall exceed 8750 lbs.

4.4.4 Hydrostatic pressure.-Prior to the application of protective finish the inside and outside tubes shall be mounted in the test device and subjected to an internal hydrostatic pressure to determine compliance with the requirements of 3.4. The test device used shall internally seal the tubes at the prescribed seal surfaces without applying an axial load other than that applied by the liquid pressure.

4.4.5 Protective finish.-Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:02Z
Check the source to verify that this is the currentversion before use.

4.4.6 Manual function.-The inside tube shall be extended manually with the clutch disengaged and without electric power.

4.4.7 Reversibility load.-The test shall be performed at midpoint of stroke with the load in tension and compression as specified in paragraph 3.6.

4.4.8 Operating loads.-The remover shall be secured in a fixture between remover mounting points and tested in accordance with 4.3.3.8 and 4.3.3.8.1. The indicated values obtained shall be recorded and used to determine compliance with 3.7.

5. PREPARATION FOR DELIVERY

5.1 Packing and Marking.-Packing and marking shall be as specified by the procuring activity. Level A - Drawings C8837890 and C9230161, Level B Drawings C8837890 and C9240994.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 A complete electrical operating cycle consists of a clutch engagement, extension, retraction, cinch-down and clutch disengagement.

6.4 Clutch engagement time is taken from the time the engage solenoid is energized and then de-energized, after clutch is engaged.

6.5 Clutch disengagement time is taken from energizing the disengage solenoid and releasing the cinch-down load to current shut-off at disengage solenoid.

Custodian:

Army - MU(FA)

Preparing activity:

Army - MU(FA)

Project No. 1377-A238

Source: https://assist.dla.mil -- Downloaded: 202640-23T01:02Z
Check the source to verify that this is the current version before use.

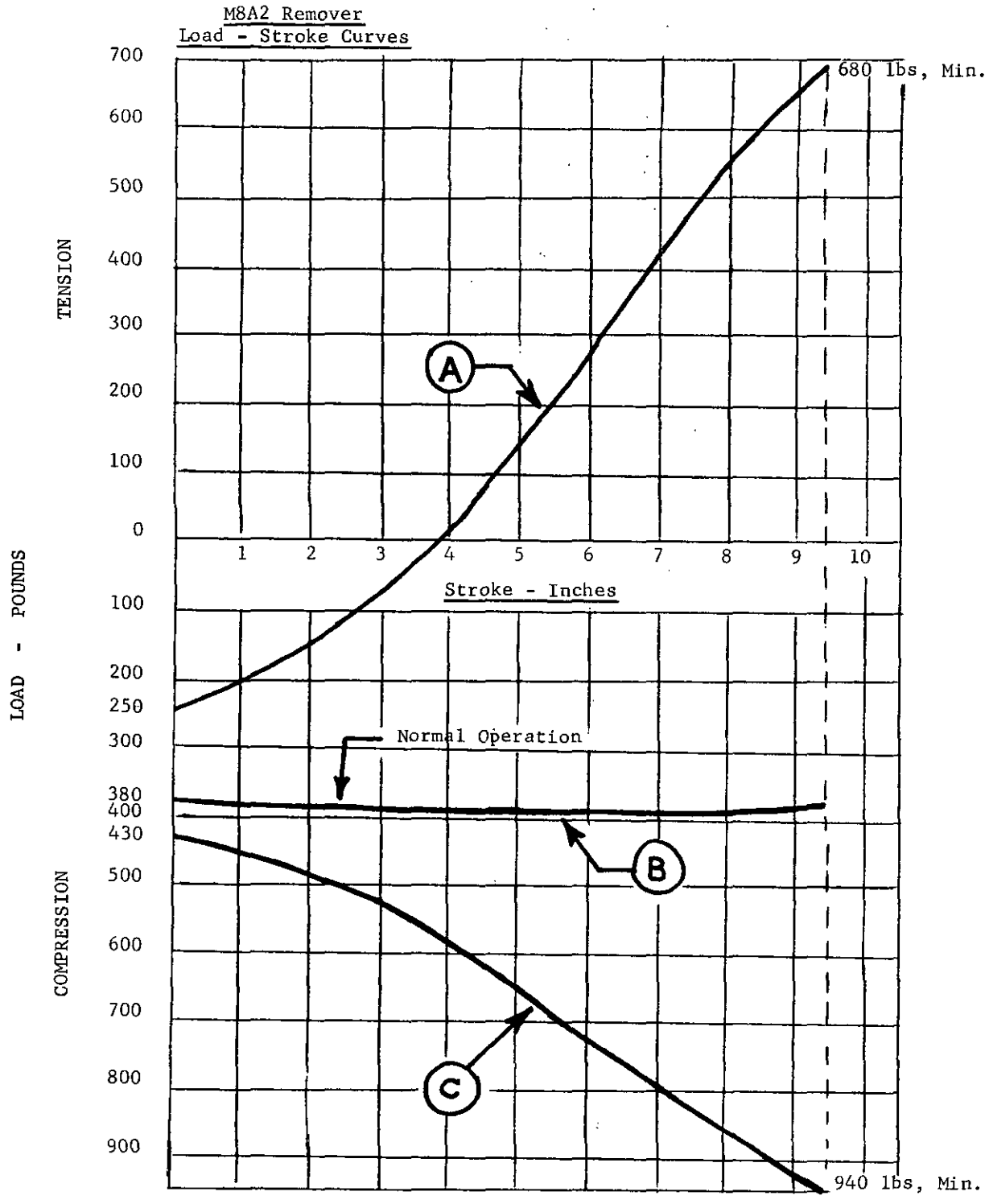


FIGURE I

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-J4000
Philadelphia, Penna. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-R-60875(MU) Remover, Aircraft Canopy, M8A2 Metal Parts Assembly

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:02Z
Check the source to verify that this is the current version before use.